Upper Rhine Robotic Mobile Challenge
Edition 2026
Règlement

September 2026

Introduction

The Upper Rhine Robotic Mobile Challenge (URMRC) is a competition open to Bachelor's and Master's level students from technological programs across Europe. It has three main objectives. First, preparing for the challenge serves as a motivating factor for students, allowing them to practically apply the technical knowledge and skills they have learned, and it can also support their academic training. Second, by promoting a positive image, this event aims to highlight scientific and technological programs dedicated to robotics during the BE 5.0 trade show. Lastly, the competition provides a valuable opportunity for interaction and camaraderie between students and industry professionals, fostering strong ties between academia and businesses.
This contest rules give the conditions for participating with the following objectives :

- Allow participation at minimal financial cost
- Showcase with high-performing robots.
- Demonstrate the expertise of academic programs.
- Promote fair play during the event.
- Simplify the jury arbitration.
- Ensure fairness among the different teams.

1 First rule- rules limits – Modification

The organizing committee reserves the right to amend these rules at any time during the competition if it deems it necessary to uphold the spirit of this event. In such a case, a meeting of the teams still in the competition will be held to inform all participants, and the changes will be posted.

Sovereign Jury: The jury has final authority, and its decisions are final. In particular, it may decide to penalize a robot and/or a team that exhibits behavior contrary to the spirit of this competition, even if the alleged offense is not explicitly covered by these rules. There is no video review. No protests based on video recordings will be considered.

2 Second rule – behavior during the event

2.1 Students behavior

The students represent their university/school/college in a public event. As such, a professional attitude is expected: a clean and organized booth, and respectful behavior. Alcohol is prohibited at the event. It is also important to respect the venue and maintain the same behavior outside the event. Any violation of this rule will result in the immediate expulsion of the students involved.

2.2 Tutor behavior

This is a student competition. Tutors accompanying a team are STRICTLY FORBIDDEN from being at their team's booth to assist in the setup or adjustment of the robots. The proper role of a tutor is to serve as part of the jury. However, they may assist students from their team or another team by suggesting improvements, but under no circumstances are they allowed to directly work on the robot or computers.

3 Third rule - Robot description

The robot is a Turtlebot3 burger developped by ROBOTIS [*](see Figure 1).

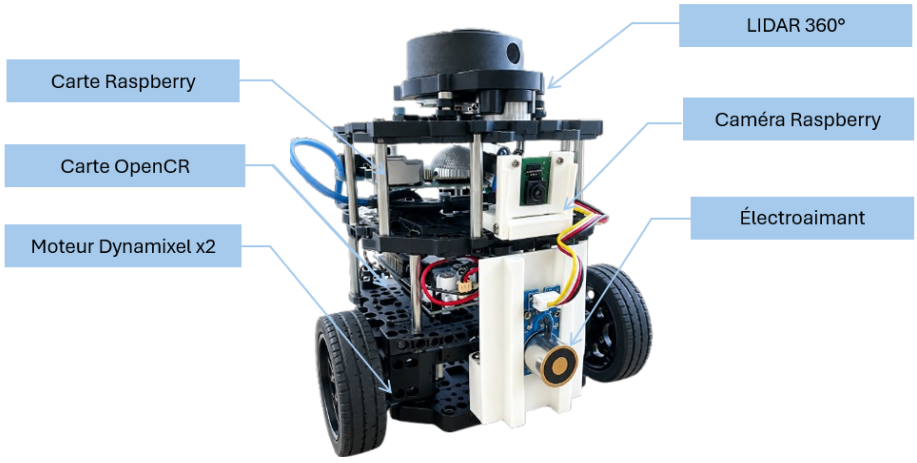


Figure 1: Turtlebot3 burger robot

It measures 138 x 178 x 192 mm and weighs approximately 1 kg. Its maximum linear speed is 0.22 m/s, and its maximum rotational speed is 162.72°/s. In terms of perception, it is equipped with a 360° LIDAR, a Raspberry Pi color camera mounted on a bracket with adjustable lateral angle, and an IMU. The control system is managed by a Raspberry Pi 4 or 3 and an OpenCR board powered by an STM32 processor. Two DYNAMIXEL motors ensure its mobility (see Figure 1), and a Li-Po battery provides energy autonomy of approximately 2.5 hours. As part of this competition, the robot is also equipped with a GROVE Electromagnet module, capable of generating a magnetic field when powered, allowing it to attract and hold small ferromagnetic metal objects.

*https://emanual.robotis.com/docs/en/platform/turtlebot3/overview/

Software development framework is ROS (Robot Operating System) or ROS2. The Signle Board Computer (SBC) Raspberry embedded a Linux Ubuntu operationg system whith a serial commmunication to OpenCR board. Programmation langages are Python, C++ or MAtlab. Robot have to be autonomous but a part of the control can be decentralized.

The robots are all identical and will remain so throughout the event. It is prohibited to modify their structure or add new equipment. Violating this rule will result in the immediate disqualification of the team. The ROS middlewar is mandatory but development language is free.

4 Fourth rule - Robot missions

All of the missions are related to issues encountered in the field of logistics with AMR-type robots Autonomous Mobile Robot). Robots have to drive on a track with a width around 20 cm. The track is in wood and bounded by two yellow and white line with a width around 2 cm (voir Figure 2). The track is on Les robots doivent rouler sur une piste de 20 cm de large en matériau Mdf délimitée par deux bandes de couleur blanche et jaune de largeur 2 cm The track is set on a 4m x 3m platfoem covered with a black carpeting.

The track consists on curves, straight lines, obstacles area, parking area, a closed area. The missions are described bellow.

4.1 Track-Following Missions

As part of a logistics mission, the robot may be required to travel from one point to another by following a predefined path, represented here by a track.

4.1.1 Track Following

In this task, perception is achieved through image processing using the data provided by the onboard camera. The mission is divided into two parts:

1. Follow a straight track over a distance of approximately 1.5 m. The track is bounded by a white line on the right and a yellow line on the left along its entire length. Stop when crossing a transverse black-and-white checkered line marking the end of the track. The robot starts from a starting line represented by a white circle centered on the track.

2. Follow a track consisting of a straight section followed by a curve. The robot stops upon detecting a STOP traffic sign (see Figure 6), which measures 5×5 cm.

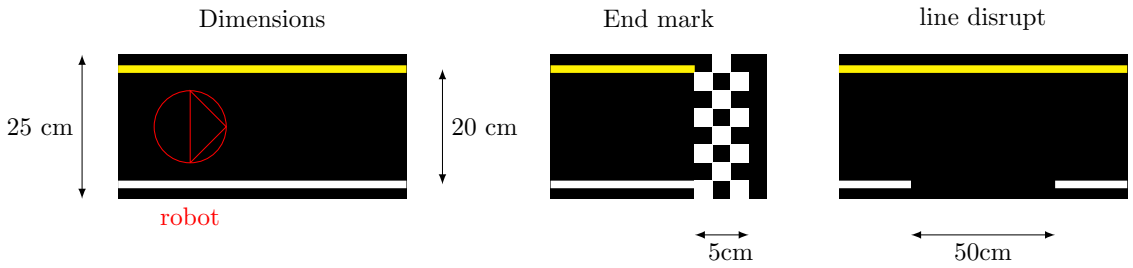


Figure 2: Track dimension and others

4.1.2 Track Following with Barrier Detection

In a warehouse, multiple robots may share the same workspace. To regulate robot traffic, barriers are placed along the track to block the passage. The robot must therefore be able to detect the barrier, stop in front of it, and remain stationary until the barrier opens (see Figure 3).

4.1.3 Track Following with an Intersection and the Right-of-Way Rule

Another approach to managing robot traffic in the warehouse is to define and enforce right-of-way rules. At an intersection, as illustrated in Figure 3, when two robots arrive simultaneously, the right-hand priority rule applies. In the example shown in Figure 3, Robot 2 has the right of way over Robot 1.

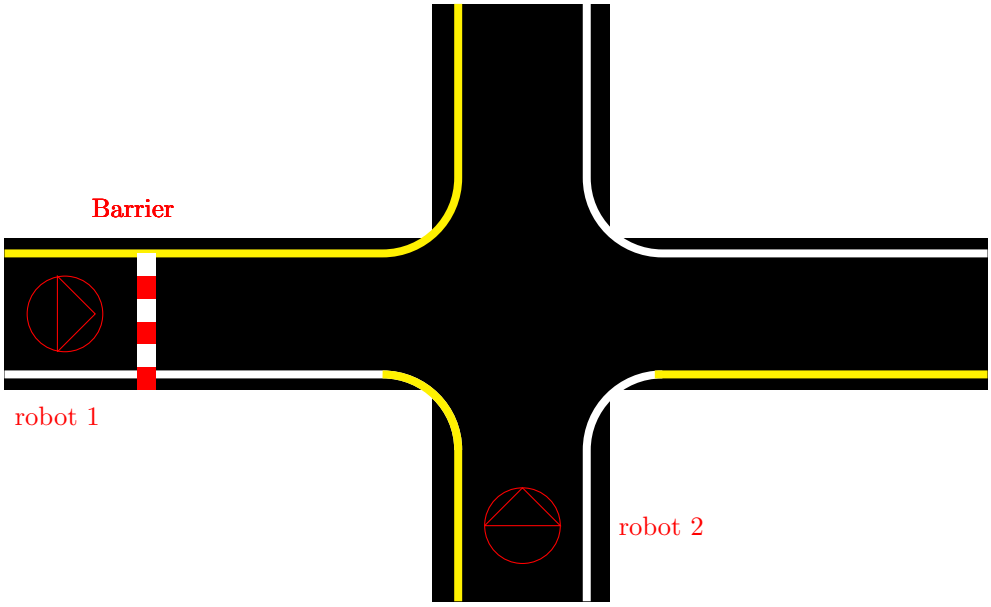


Figure 3: Representation of the intersection area.

4.2 Missions Related to Obstacle Avoidance and Maneuvering

In a warehouse environment, obstacles are likely to be encountered frequently and under varying lighting conditions. When illumination is sufficient, an onboard camera can detect obstacles effectively. However, some areas may be dimly lit or completely dark, significantly reducing the quality of the visual information available.

4.2.1 Navigating and Exiting an Enclosure

In this mission, the robot must navigate through a poorly lit enclosure using only LiDAR data for environmental perception. The enclosure is a confined area bounded by walls, with one opening serving as the entrance and a second opening serving as the exit (see Figure 4). Several obstacles are placed randomly inside the enclosure.

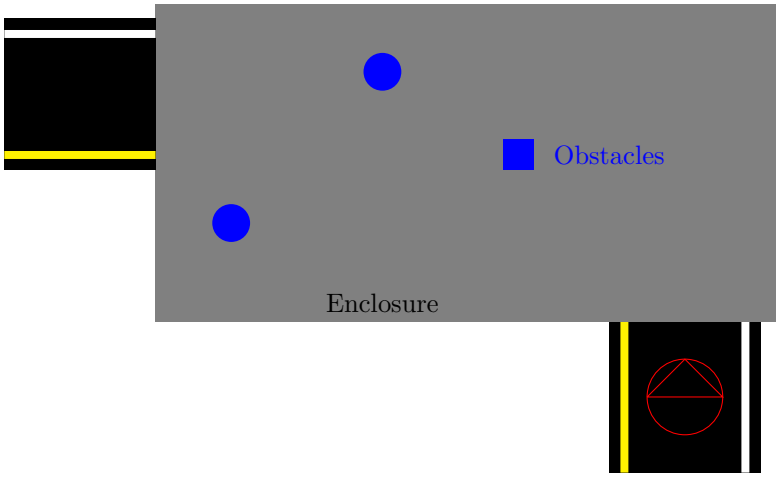


Figure 4: Enclosure without the use of a camera.

The mission is divided into three parts:

1. The robot must detect the entrance opening and enter the enclosure without colliding with the doorway.
2. Once inside, the robot must locate the exit and leave the enclosure without touching the walls within a time limit of 1 minute.
3. Finally, the robot must repeat the previous two tasks with randomly placed obstacles inside the enclosure.

During this mission, the use of the camera is strictly prohibited.

4.2.2 Storage and Retrieval Maneuver

The storage/retrieval area contains three distinct zones numbered ① to ③, identified by traffic signs placed behind each zone. The positions of these numbered zones are assigned randomly. Zone ① is initially occupied by a pallet measuring 15 cm × 15 cm × 5 cm, resting on a 10 cm-high support. The front side of the pallet includes a metallic plate that allows the robot to grasp it using an electromagnet (e.g., a Grove electromagnet).

The objective is to move the pallet from Zone ① to Zone ②. The mission is divided into four parts:

1. Detect the zone numbered ① containing the pallet.
2. Grasp the pallet using the electromagnet.
3. Transport the pallet from Zone ① to Zone ② without crossing any other storage zones and while preserving its initial orientation.
4. Release the pallet at the center of Zone ② and exit the storage area.

The electromagnet control is left free. Four landmarks (fixed reference points that can be identified without ambiguity and are used for navigation) positioned at the four corners of the area can be used for localization.

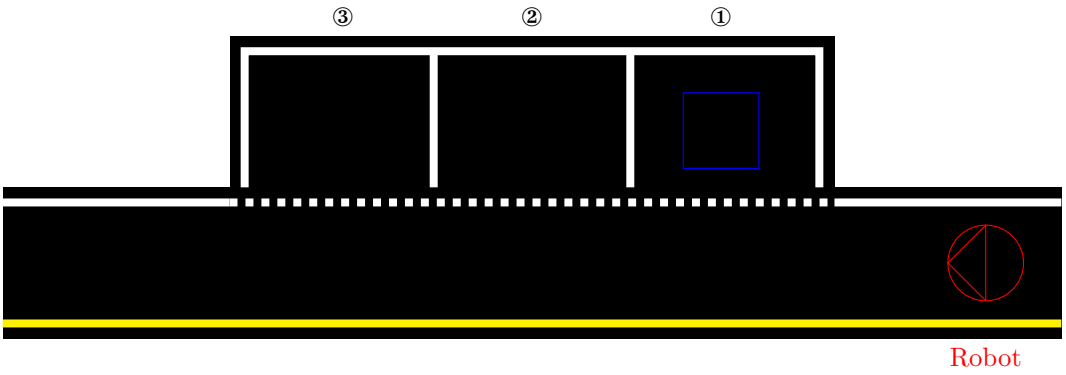


Figure 5: Storage and retrieval area



Figure 6: Set of traffic signs used on the challenge field

5 Challenge Organization

The competition is organized into five phases: registration, development, qualification, challenge, and award ceremony.

5.1 Registration

To participate in the competition, teams must submit a pre-registration to the organizers by the dates indicated on the competition website, specifying the number of teams and the names of the supervising instructors. This pre-registration is then confirmed in September with the names of the three student members of the team, their email addresses, the name of their institution, and the names of the supervising instructors responsible for the team. Upon arrival at the venue, the team must report to the organizers at the specified dates and times in order to provide the name chosen for the robot.

5.2 Development

Following registration, students will have access to a simulation environment (see Figure 7) including:

- Example of the track layout,

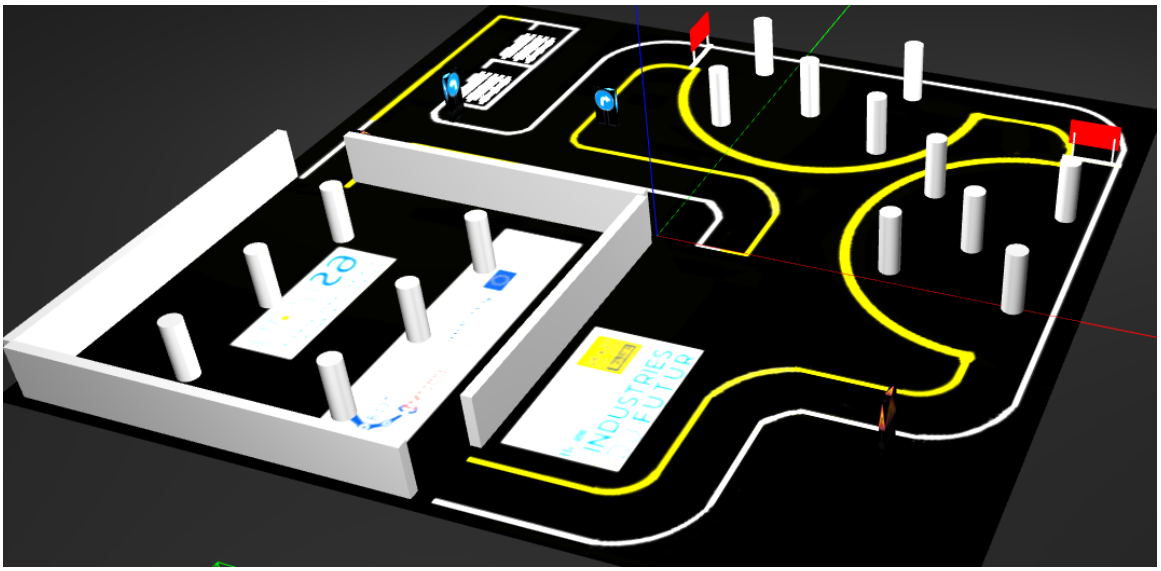


Figure 7: Illustration of the robot challenge environment in Gazebo

- Simulation model of the robot,
- Examples of missions.

This environment allows students to develop and test their various solutions for carrying out the missions described in Section 4. The use of the **ROS** or **ROS2** framework is mandatory. The programming language is left to the participants' choice.

Each team must then have access to a robot no later than two weeks after the launch of the competition. This will allow teams to test the various components, develop the missions directly on the physical robot, and prepare for the qualification stage.

5.3 Qualification

Two weeks after the launch of the challenge, each team will be invited to a videoconference with the organizing team. During this first meeting, each team will have the opportunity to ask questions and request any necessary clarification regarding the challenge procedure, rules, and various missions.

One month later, each team will also be invited to a second videoconference with the organizing team. This meeting will be dedicated to the qualification process, during which each team will be required to demonstrate that its robot is functioning properly and successfully complete the track-following mission.

5.4 Challenge

The challenge takes place over the two days of the BE 5.0 trade fair. The different teams will be welcomed during the first half-day.

The missions must be completed according to the following schedule:

- Day 1: Missions related to track following

- Day 2: Missions related to obstacle avoidance and maneuvering

In order to test their applications, teams will be allocated 15-minute time slots during which they will have access to the track to test and validate their results. Sections of the track will also be made available at the booths for testing purposes. However, depending on track availability, missions may be tested independently of the established order at the request of the teams and with the approval of the judging panel.

At the end of each day, all teams will be invited to participate in a collective challenge.

5.5 Points attribution

To establish a ranking, points are awarded according to the distribution shown in Table 1.

5.6 Awards ceremony

The challenge is closed by the awards ceremony for the first four teams. It is planned on the 2ème day from 15h30 with all the teams and officials.

Missions	No.	Description	Points	Bonus	Penalty
Track-following missions	0	Track following + stopping at the checkered line	4		-1 (if the robot leaves the track or fails to stop)
		Correct track following through the curve + stopping at the STOP sign	1		-1 (if the robot leaves the track or fails to stop)
	1	Barrier detection and stopping the robot in front of the barrier	3		-1 (if the robot hits the barrier)
		Track following after the barrier opens	2		
	2	Stopping the robot after detecting the robot on the right	3		-1 (if no detection occurs)
		Track following with right-of-way priority	2		-2 (if the robot takes the wrong direction)
	3	Team challenge	8		-1 per finishing position (timed)
Obstacle avoidance and maneuvering missions	4	Detection of the tunnel entrance sign	1		
		Robot entering the enclosure	3		-1 (if a wall is touched)
		Robot exiting the enclosure	3		-1 (if a wall is touched)
		With obstacle		+3	-1 (if an obstacle falls)
	5	Detection of the occupied pallet zone and correct positioning of the robot		+2	-1 (if the robot fails to stop)
		Pallet pickup	1		
		Pallet transportation	3		-1 if another zone is crossed
		Pallet release	1	+1 if the robot leaves the zone	
	6	Team challenge	8		-1 per finishing position (timed)

Table 1: Point distribution.

Article 6 - Photo right

As part of the competition, photos and videos are produced and processed. Their purpose is to document the activities of the competition. The collected data is electronically saved and processed. For example, it may be published on the organizers' website, the participating institutions' websites, their social media, flyers, posters, etc. Some photos or videos may also be shared with the INTERREG Program managing authority, sponsors, or cooperation partners solely for the purpose of documenting the competition.



URMRC 2024 edition



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